



GOA UNIVERSITY

B.E - (Computer) (Sem-VII)(Revised Course 2019-2020)

Paper Title: Compiler Design

Paper Code: CE 710

[Duration: 3:00 Hours]

[Total Marks: 100]

Instructions: Answer any five questions by selecting 2 questions each from **part A & part B** & 1 question from **part C**.

Part A

- Q1 a. Define a Compiler. Show the output of each phase of the compilation for the following code: 8
- ```

if(i<10)
{
 i=i+1;
}
else
{
 i=i+10;
}

```
- b. Explain the concept of bootstrapping and porting with suitable examples. 6
- c. Write a short note on Lex. Write a Lex program to count the number of vowels and consonants in a string. Define YYWRAP function in the program and show the commands of execution. 6
- Q2 a. Define context free grammar for the following statements. 6
- (a) for loop with at least one if statement inside for loop.
- (b) if statement with one assignment statement inside if
- b. Write context free grammar for  $L = \{a^n b^{n+1} \mid n \geq 1\}$ . Using LL(1) parsing, parse the string: aaabbb 8

- c. Define Left Recursion. Eliminate left recursion from the grammar given below: 6  
**Stmt**  $\rightarrow$  **if(E) {S1} else {S2}**  
**E**  $\rightarrow$  **id relop id**  
**S1**  $\rightarrow$  **id = expr**  
**expr**  $\rightarrow$  **expr + expr | expr - expr | id**

- Q3 a. Write short note on YACC. State the commands used in execution of YACC program. 6

- b. Compute first and follow for the grammars given below: 6  
**i) S**  $\rightarrow$  **PQR**  
**P**  $\rightarrow$  **g**  
**Q**  $\rightarrow$  **ap | €**

- ii) exp**  $\rightarrow$  **void func\_name(B)**  
**B**  $\rightarrow$  **id, B | id**  
 c. Explain 2 Pass Assembler with a neat diagram. 8

### Part B

- Q4 a. Write syntax directed translation scheme for switch statement with suitable example. 6

- b. Construct the canonical collection of LR(0) items and SLR(1) parsing table for the following grammar: 8  
**S**  $\rightarrow$  **ABy**  
**A**  $\rightarrow$  **Sy | x**  
**B**  $\rightarrow$  **y S**

- c. Construct Operator precedence relation table for the grammar given 6  
**S**  $\rightarrow$  **a A c B e**  
**A**  $\rightarrow$  **Ab | d**  
**B**  $\rightarrow$  **d**

- Q5 a. Define quadruples, triples and indirect triples representation. For the expressions given below, give all three representations: 8  
**A** = **K[i] \* D[i] + d/2**  
**D** = **-a \* c + -a \* c / d + q / r**

- b. Explain code generation algorithm. Generate code for the expression given below. Also show the contents of register and address descriptor at every step. 8  
**a = b + a \* a + c / d + z / b**

- c. What are the applications of dags? Construct a dag for: 4  
**a = b + c**  
**b = b - d**  
**c = c + d**  
**e = b + d**

- Q6 a. What is peephole optimization? Discuss any three peephole optimization techniques with the help of examples. 6

- b. Explain different Error recovery strategies. 6
- c. Compute set of LALR items and show parsing table. Are there any conflicts in the parsing table? 8  
 $S \rightarrow X X$   
 $X \rightarrow a X$   
 $X \rightarrow b$

**Part C**

- Q7 a. List and explain different issues in design of code generator. 8
- b. Explain the following advanced assembler directives with suitable examples: 6
  - i. LTORG
  - ii. EQU
  - iii. ORIGIN
- c. Define backpatching. Using backpatching construct parse tree for  $A > B$  or  $C > D$  and  $P < Q$  6  
 Use appropriate translation scheme in this problem.
- Q8 a. Write a short note on different compiler construction tools. 6
- b. Write syntax directed translation scheme for translating Boolean expressions using numerical representation. Using this SDT, generate code for : 8  
 $p \text{ AND } q \text{ OR NOT } s$
- c. Define Tokens, Lexeme and Pattern. Discuss the role of finite automata in Lexical analysis. 6